

Answer all questions

1. For the network shown in Fig. 1, $R_1 = 600\text{K}\Omega$, $R_2 = 400\text{K}\Omega$ and $C = 0.2\mu\text{F}$. Determine the maximum phase provided by this network and the frequency (in Hz) at which this phase occurs. [6]

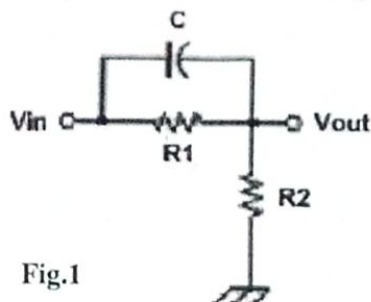


Fig.1

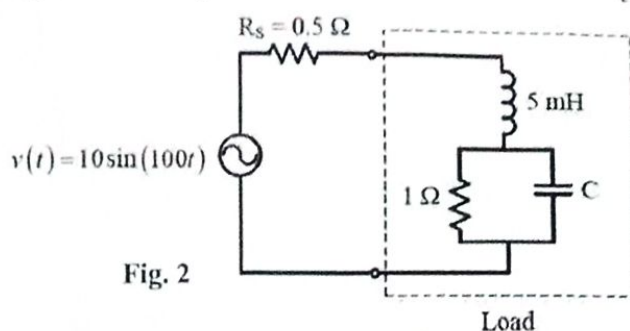


Fig. 2

2. In the circuit shown in Fig. 2, determine the value of the capacitor required for maximum power to be transferred to the load. [3]
3. The network $\mathcal{N}$ shown in Fig. 3a and 3b is linear, bilateral and passive. Determine the current I in Fig. 3b. [7]

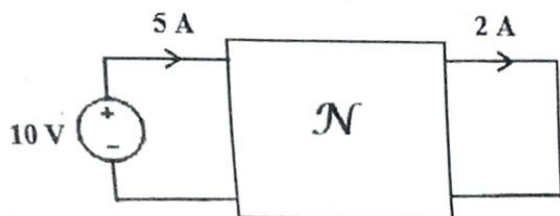


Fig. 3a

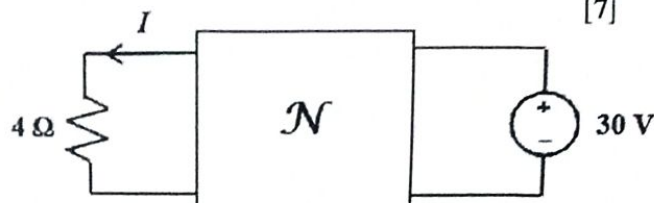


Fig. 3b

4. The switch in the circuit in Fig. 4 has been open a long time before closing at $t = 0$. Find $i_L(t)$ for $t \geq 0$. [9]

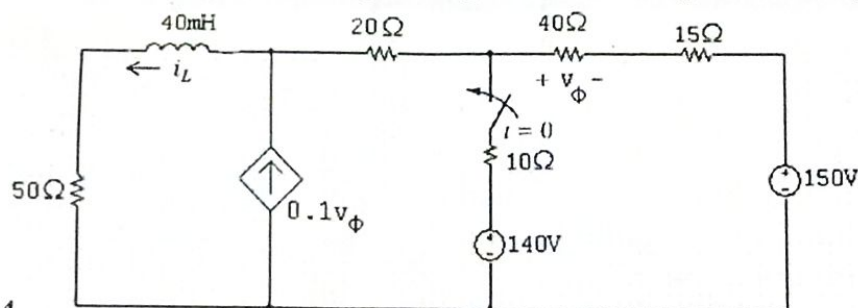


Fig. 4

5. Draw appropriate graph and obtain the topological formulation of loop equations for the network shown in Fig. 5. Follow the convention studied in the class for drawing the graph and the formulation of loop equations. Take 30Ω and 40Ω branches as tree branches. [7]

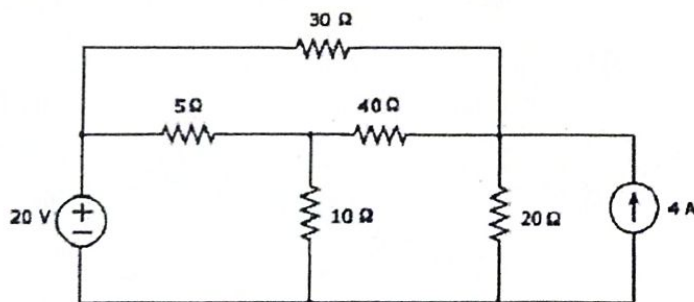


Fig. 5

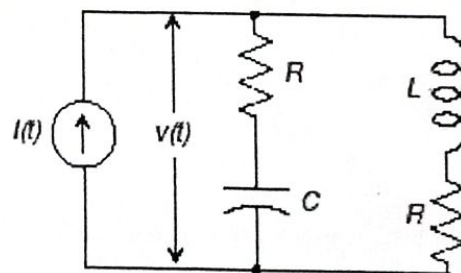


Fig. 6

6. In the circuit shown in Fig. 6, the input is $I(t)$ and the output is $V(t)$.
- Obtain the state variable model of the circuit.
 - Find a relationship between R , L , and C that renders the system uncontrollable.
 - Determine the network function relating $V(s)$ to $I(s)$ when the system is uncontrollable.
 - Is the system observable when it is uncontrollable?

[8]